Cross-Cohort Validation of Clinical Prediction Models in First-Episode Psychosis

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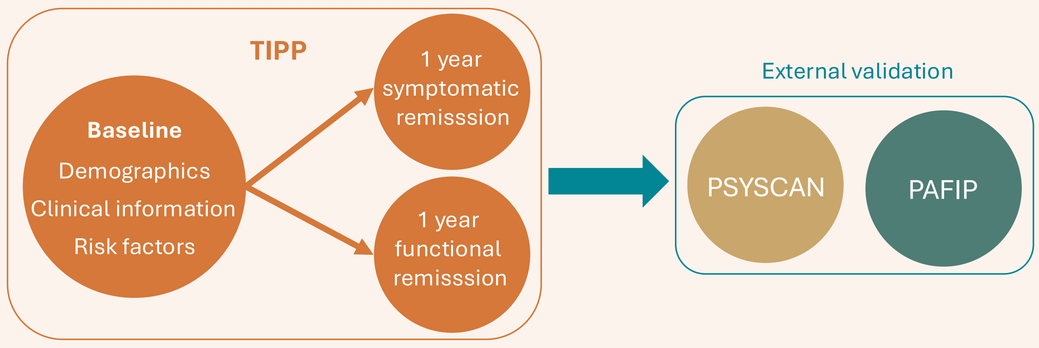
Introduction

First-episode psychosis trajectories are heterogeneous, and it remains difficult to predict who will have favourable or unfavourable outcomes [ref].

Several prediction models have been developed to estimate likely treatment outcomes, but very few have been externally validated [review]. External validation is essential because it provides a more rigorous test of whether the information learned by a model truly distinguishes between patients who later improve and those who do not [ref]. This remains a major challenge, as clinical services often differ in procedures, assessments, and patient characteristics, creating inherent differences across samples. Addressing this issue is critical for the development of clinically useful models.

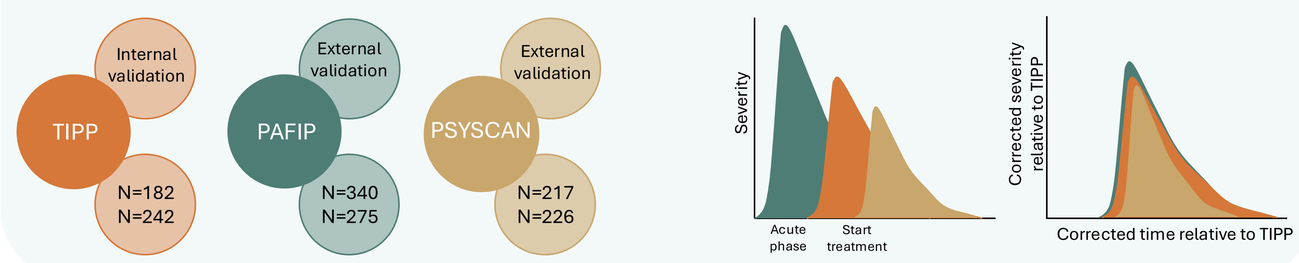
Previous studies have used a range of predictors, including clinical, imaging, and omics data [review]. Routinely acquired clinical data may be the most promising for implementation, as it is already available in practice and does not require additional costly assessments.

Aim

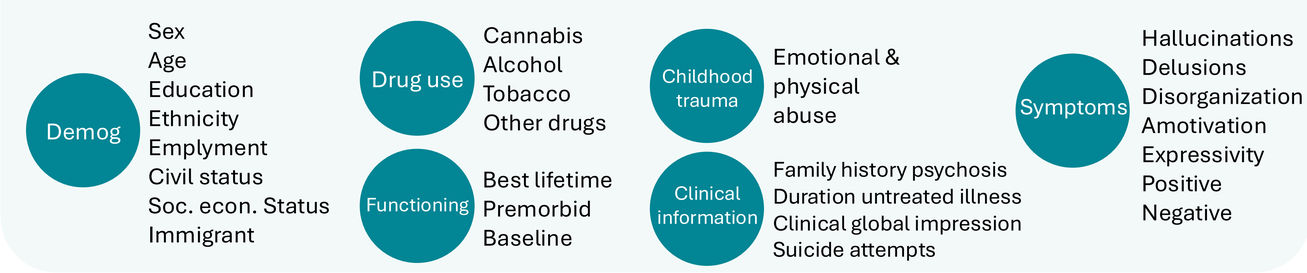


Methods

Datasets

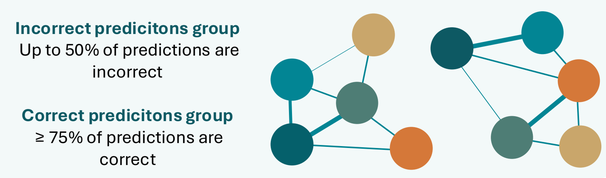


MEASURES & HARMONISATION



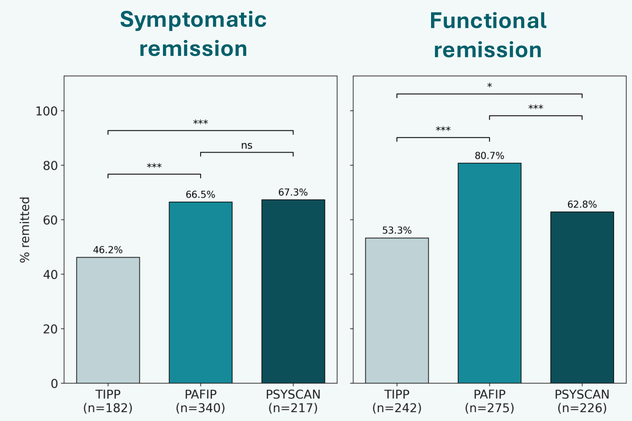
Machine learning

Model bias



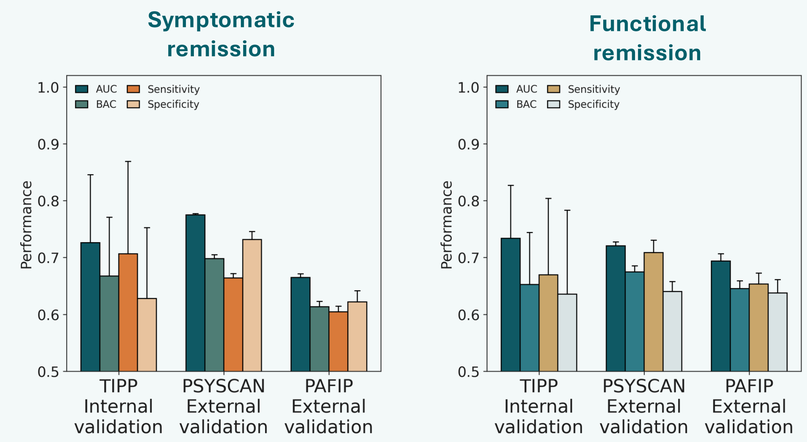
Results

1. Sample charateristics



	Symptomatic remission			Functional remission		
	TIPP vs PAFIP	TIPP vs PSYSCAN	PAFIP vs PSYSCAN	TIPP vs PAFIP	TIPP vs PSYSCAN	PAFIP vs PSYSCAN
Sex	0.19	-0.04	-0.23*	0.17	-0.12	-0.29**
Civil status	-0.26*	-0.11	0.16	-0.30**	-0.07	0.23*
Employment status	-0.09	0.06	0.15	-0.08	0.05	0.13
Student	0.07	0.32**	0.24**	0.14	0.34***	0.19
Immigrant	-0.74***	-0.42***	0.31***	-0.83***	-0.42***	0.41***
Ethnicity	-0.62***	0.14	0.75***	-0.73***	0.05	0.78***
Socioeconomic status	0.96***	0.01	-0.96***	1.09***	0.10	-0.99***
Winter birth	-0.03	0.09	0.13	-0.05	0.01	0.06
Family history psychosis	0.09	-0.31**	-0.40***	0.17	-0.25*	-0.43***
Cannabis use	0.06	0.04	-0.01	0.07	0.06	-0.01
Tobacco use	-0.08	0.14	0.22*	-0.01	0.21*	0.22*
Alcohol use	-0.35***	0.19	0.55***	-0.39***	0.18	0.57***
Childhood abuse	-0.66***	-0.16	0.49***	-0.53***	-0.11	0.43***
Age	0.75***	-0.02	-0.76***	0.84***	-0.03	-0.83***
Years of education	-0.09*	1.02***	1.12***	-0.22***	0.93***	1.13***
Clinical Global Impression	1.83***	-0.80***	-2.87***	1.83***	-0.72***	-2.73***
Duration of untreated psychosis	-0.05	-0.26**	-0.18***	-0.09	-0.26***	-0.20***
Number of suicide attempts	-0.37**	0.22	0.46***	-0.33**	0.20	0.42***
Functioning best lifetime	1.24***	0.83***	-0.48***	1.20***	0.81***	-0.45***
Functioning baseline	0.35	0.63***	0.08	0.33	0.58***	0.07
Other drug use	0.19	0.16	-0.03	0.15*	0.12	-0.03
Premorbid adjustment	0.37***	-0.09	-0.46***	0.33**	-0.07	-0.41***
Delusions	3.74***	-0.06	-3.07***	3.29***	-0.03	-2.79***
Disorganization	1.75***	0.08	-1.65***	1.91***	0.13	-1.76***
Hallucinations	0.88***	0.06	-0.83***	0.94***	0.12	-0.81***
Expressivity	-0.66***	-0.30**	0.34***	-0.59***	-0.20*	0.39***
Amotivation	-0.93***	-0.48***	0.51***	-0.84***	-0.36**	0.53***
Positive symptoms	2.52***	-0.01	-2.25***	2.58***	0.05	-2.20***
Negative symptoms	-0.64***	-0.43***	0.19**	-0.51***	-0.30**	0.21**

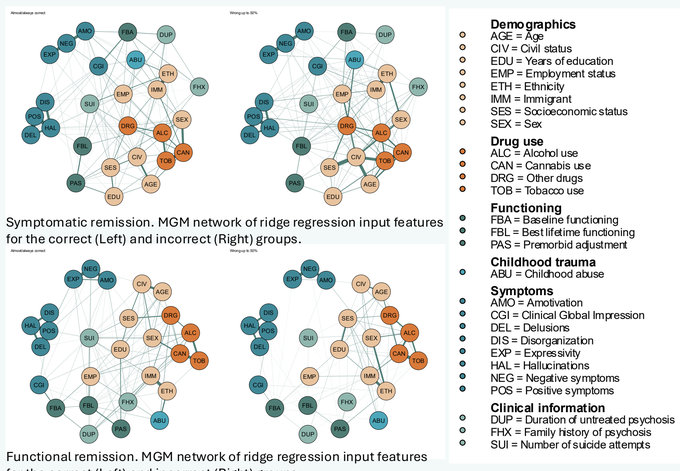
2. Model performance and most informative features



3. Model bias

	Group (n)	Total n (%)	TIPP n (%)	PSYSCAN n (%)	PAFIP n (%)	% of features sig ≠
Symptomatic remission	Correct	459 (63.7)	116 (65.2)	145 (68.3)	198 (59.8)	22.2% inc > corr symptoms
	Incorrect	262 (36.3)	62 (34.8)	67 (32.6)	133 (40.2)	
Functional remission	Correct	455 (63.1)	150 (62.8)	144 (66.1)	161 (62.4)	0%
	Incorrect	255 (35.9)	89 (37.2)	74 (33.9)	97 (37.6)	

Size and univariate comparison between the Correct and Incorrect prediction groups on models' input features adjuted for cohort and multiple comparisons.



Symptomatic remission. MGM network of ridge regression input features for the correct (Left) and incorrect (Right) groups.

Functional remission. MGM network of ridge regression input features for the correct (Left) and incorrect (Right) groups.

Outcome	Network structure invariance	Global strength invariance	Strength correlation stability
Symptomatic remission			
Functional remission	M = 0.5 (p=0.175)	S = 3.1 (p = 0.299)	CS_{cor} = 0.67 CS_{inc} = 0.60

Comparison of network between the Correct and Incorrect predictions groups.

Conclusion

- External validation remains rare in FEP prediction research → this study provides validation across two independent cohorts.
- Models trained in TIPP showed good transportability to both external datasets, despite expected differences between services and samples. For symptomatic remission, the model achieved an AUC of 0.72 in internal validation and 0.78 and 0.66 in external. For functional remission, the model achieved an AUC of 0.73 in internal validation and 0.72 and 0.69 in external validation.
- Symptomatic remission was driven more by symptom severity, whereas functional remission was driven by psychosocial and functioning-related variables.
- There were no differences in clinical profiles between patients with correct and incorrect predictions for functional remission, suggest the model has no apparent bias.
- For symptomatic remission, patients with more severe symptoms had lower rate of correct predictions -> bias towards less severe patients.